

Nonlinear Spectroscopy II: Phase Matching Condition and Double-Sided Feynman Diagrams

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1 Phase Matching Condition

The optical process in a nonlinear material can be described using classical electrodynamics. The Maxwell equation in a medium is (in SI units)

$$\begin{aligned}\nabla \cdot \mathbf{D} &= \rho_f \\ \nabla \times \mathbf{H} &= \mathbf{j}_f + \frac{\partial \mathbf{D}}{\partial t} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}\end{aligned}\tag{1}$$

where ρ_f and $\mathbf{j}_f$ are free charge density and free current density. Here we only consider non-magnetic material, i.e., $\mathbf{B} = \mu_0 \mathbf{H}$. The material is nonlinear in the sense that $\mathbf{D}$ is not linear in $\mathbf{E}$, but also includes higher-order terms. We write $\mathbf{D}$ as

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}\tag{2}$$

where $\mathbf{P}$ is the induced polarization density. When there is no free charge in space, the wave equation of electromagnetic field in nonlinear material can be derived from the Maxwell equation,

$$\nabla \times \nabla \times \mathbf{E}(\mathbf{r}, t) + \frac{1}{c^2} \frac{\partial^2 \mathbf{E}(\mathbf{r}, t)}{\partial t^2} = -\frac{1}{\varepsilon_0 c^2} \frac{\partial^2 \mathbf{P}(\mathbf{r}, t)}{\partial t^2}\tag{3}$$

In this equation, $\mathbf{P} = \mathbf{P}^{(1)} + \mathbf{P}_{\text{NL}}$ contains both the linear polarization $\mathbf{P}^{(1)}$ and the nonlinear polarization $\mathbf{P}_{\text{NL}}$. For the linear polarization, we have

$$\mathbf{P}^{(1)}(\mathbf{r}, t) = \int d\mathbf{r}' \int dt' \mathcal{R}^{(1)}(\mathbf{r} - \mathbf{r}', t - t') \mathbf{E}(\mathbf{r}', t')\tag{4}$$

where $\mathcal{R}^{(1)}$ is the linear response function. The dielectric function (a rank-2 tensor) is defined as

$$\varepsilon(\mathbf{r} - \mathbf{r}', t - t') = \delta(\mathbf{r} - \mathbf{r}') \delta(t - t') + \frac{1}{\varepsilon_0} \mathcal{R}^{(1)}(\mathbf{r} - \mathbf{r}', t - t')\tag{5}$$

The wave equation can thus be written as

$$\boxed{\nabla \times \nabla \times \mathbf{E}(\mathbf{r}, t) + \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \int d\mathbf{r}' \int dt' \varepsilon(\mathbf{r} - \mathbf{r}', t - t') \mathbf{E}(\mathbf{r}', t') = -\frac{1}{\varepsilon_0 c^2} \frac{\partial^2 \mathbf{P}_{\text{NL}}(\mathbf{r}, t)}{\partial t^2}}\tag{6}$$

1.1 Linear Optics

The nonlinear wave equation is an inhomogeneous differential equation, the homogeneous part is a linear differential equation of $\mathbf{E}$,

$$\nabla \times \nabla \times \mathbf{E}(\mathbf{r}, t) + \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \int d\mathbf{r}' \int dt' \varepsilon(\mathbf{r} - \mathbf{r}', t - t') \mathbf{E}(\mathbf{r}', t') = 0 \quad (7)$$

This equation describes linear optical processes such as absorption. Using the relation

$$\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \quad (8)$$

and the fact that in linear medium with no free charge, $\nabla \cdot \mathbf{E} = 0$ because $\nabla \cdot \mathbf{D} = 0$, we have

$$\nabla^2 \mathbf{E}(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \int d\mathbf{r}' \int dt' \varepsilon(\mathbf{r} - \mathbf{r}', t - t') \mathbf{E}(\mathbf{r}', t') = 0 \quad (9)$$

In isotropic medium where ε becomes a scalar, this equation has general solutions in the form of plane waves,

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (10)$$

and the wave vector $\mathbf{k}$ and the angular frequency ω satisfy the dispersion relation, which is obtained through Fourier transformation of Eq. (9),

$$\frac{kc}{\omega} = \sqrt{\varepsilon(\mathbf{k}, \omega)} \equiv n_r(\mathbf{k}, \omega) + i\kappa(\mathbf{k}, \omega) \quad (11)$$

$\varepsilon(\mathbf{k}, \omega)$ is the Fourier transformation of $\varepsilon(\mathbf{r}, t)$, and $n_r(\mathbf{k}, \omega)$ and $\kappa(\mathbf{k}, \omega)$ are the index of refraction and extinction coefficient, respectively. **In linear optical processes, waves with different frequencies are linearly independent solutions of the homogeneous equation. Therefore, there will be no mixing or coupling between waves with different frequencies, such mixing is a result of the nonlinear, inhomogeneous term.**

1.2 Multiwave Mixing

The nonlinear polarization $\mathbf{P}_{\text{NL}}$ carries the complete microscopic information required for describing any nonlinear optical processes. In the inhomogeneous wave equation, $\mathbf{P}_{\text{NL}}(\mathbf{r}, t)$ serves as a source that generates the signal field. In frequency domain, we can write $\mathbf{P}_{\text{NL}}(\omega)$ as power series of $\mathbf{E}(\omega)$,

$$\mathbf{P}_{\text{NL}}(\omega) = \sum_{n=2}^{\infty} \chi^{(n)}(\omega) \cdot \mathbf{E}^n(\omega) \quad (12)$$

where $\chi^{(n)}(\omega)$ is the **n th-order susceptibility**, which is a rank $n+1$ tensor. With this expansion, Eq. (6) becomes a closed nonlinear differential equation of $\mathbf{E}(\mathbf{r}, t)$, and $\mathbf{E}(\mathbf{r}, t)$ can be solved self-consistently. **However, in general the signal field is much weaker than the incoming fields. We can thus ignore the induced polarization due to the signal field and treat $\mathbf{P}_{\text{NL}}(\mathbf{r}, t)$ as a given function. In other words, $\mathbf{P}_{\text{NL}}(\mathbf{r}, t)$ depends only on the incoming**

fields, which are unaffected by the wave mixing process. In this case, Eq. (6) becomes a linear inhomogeneous equation.

In a $n + 1$ wave mixing process, n incoming waves interact with matter, generating a new field. The electric field corresponding to the incoming waves is

$$\mathbf{E}_{\text{in}}(\mathbf{r}, t) = \sum_j^n \left[\mathbf{E}_j(\mathbf{r}, t) e^{i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} + \mathbf{E}_j^*(\mathbf{r}, t) e^{-i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} \right] \quad (13)$$

where $\mathbf{E}_j(\mathbf{r}, t)$ describes the slowly varying envelope of the j th incoming wave. Using Eq. (12), $\mathbf{P}_{\text{NL}}(\mathbf{r}, t)$ should take the form

$$\boxed{\mathbf{P}_{\text{NL}}(\mathbf{r}, t) = \sum_{n=2}^{\infty} \sum_s \mathbf{P}_s^{(n)}(\mathbf{r}, t) e^{i(\mathbf{k}_s \cdot \mathbf{r} - \omega_s t)} + c.c.} \quad (14)$$

where $\mathbf{P}_s^{(n)}(\mathbf{r}, t)$ is the slowly varying envelope determined by $\mathbf{E}_j(\mathbf{r}, t)$ and $\chi^{(n)}$, $\mathbf{k}_s$ and ω_s **are any linear combinations of the incoming wave vectors and frequencies.** Take the three-wave mixing as an example and only keep the $n = 2$ term, we have

$$\omega_s = 2\omega_1, \ 2\omega_2, \ \omega_1 + \omega_2, \ \omega_1 - \omega_2, \ 0 \quad (15)$$

These processes correspond to second harmonic generation, sum frequency generation, difference frequency generation and optical rectification, respectively.

In many multiwave mixing experiments, we choose the frequency of the waves outside the absorption spectrum to avoid absorption loss. In this case, $\varepsilon(\mathbf{k}, \omega)$ is real. We further assume that the medium is isotropic and the linear response is local in space, therefore

$$\varepsilon(\mathbf{k}, \omega) = \varepsilon(\mathbf{k} = 0, \omega) = n_{\text{r}}^2(\omega) \quad (16)$$

In nonlinear medium with no free charge, $\nabla \cdot \mathbf{D} = 0$ does not necessarily imply $\nabla \cdot \mathbf{E} = 0$, because $\mathbf{D}$ also encompasses $\mathbf{E}^2$, $\mathbf{E}^3$, ... terms. However, even when $\nabla \cdot \mathbf{E}$ is not identically 0, it is often negligibly small and thus in practice is usually ignored. The wave equation for the signal field can thus be simplified as

$$\nabla^2 \mathbf{E}_{\text{sig}}(\mathbf{r}, t) - \frac{n_{\text{r}}^2}{c^2} \frac{\partial^2 \mathbf{E}_{\text{sig}}(\mathbf{r}, t)}{\partial t^2} = \frac{1}{\varepsilon_0 c^2} \frac{\partial^2 \mathbf{P}_{\text{NL}}(\mathbf{r}, t)}{\partial t^2} \quad (17)$$

Consider the following experimental setup. A film of material with thickness l ($k_s l \gg 1$, the thickness is much larger than the wave length) is placed on the xy plane. The multiwave mixing thus occurs in the region $0 < z < l$ and the signal is measured at some $z > l$. Both the incoming fields and the signal are propagating along the z axis. Therefore, Eq. (17) reduces to a 1D equation,

$$\frac{\partial^2 \mathbf{E}_{\text{sig}}(z, t)}{\partial z^2} - \frac{n_{\text{r}}^2}{c^2} \frac{\partial^2 \mathbf{E}_{\text{sig}}(z, t)}{\partial t^2} = \frac{1}{\varepsilon_0 c^2} \frac{\partial^2 \mathbf{P}_{\text{NL}}(z, t)}{\partial t^2} \quad (18)$$

For the incoming fields, we assume that there is only a temporal envelope, i.e., $\mathbf{E}_j(t)$ and $\mathbf{P}_s^{(n)}(t)$ are independent of z . We focus on one Fourier component of $\mathbf{P}_{\text{NL}}(z, t)$,

$$\mathbf{P}_{\text{NL}}(z, t) = \mathbf{P}_s(t) e^{i(k_s z - \omega_s t)} + c.c. \quad (19)$$

The emitted electromagnetic wave should have the same frequency as its source, we thus look for solutions with the form

$$\mathbf{E}_{\text{sig}}(z, t) = \mathbf{E}_s(z, t)e^{i(k'_s z - \omega_s t)} + c.c. \quad (20)$$

where

$$k'_s = \frac{n_s \omega_s}{c} \quad (21)$$

and $n_s \equiv n_r(\omega_s)$. The envelope $\mathbf{E}_s(z, t)$ depends on z because the signal grows from zero at $z = 0$ and intensifies as it travels through the sample.

Since the index of refraction is frequency-dependent, generally,

$$\boxed{k_s = \sum_j' k_j = \sum_j' \frac{n_j \omega_j}{c} \neq \frac{n_s}{c} \sum_j' \omega_j = k'_s} \quad (22)$$

because $n_j \equiv n_r(\omega_j) \neq n_r(\omega_s) \equiv n_s$. Here $\sum_j'$ represents a certain linear combination of k_j or ω_j .

$$\Delta k = k_s - k'_s \quad (23)$$

is called the **wave vector mismatch**. In this collinear setup, the wave vector mismatch is caused by the frequency-dependence of the index of refraction. In a non-collinear setup, another cause of wave vector mismatch is that k_s is obtained by vector addition of k_j whereas k'_s is obtained by scalar addition of ω_j and the dispersion relation.

Substituting Eqs. (19) and (20) into Eq. (18), we can get a differential equation for the envelope $\mathbf{E}_s(z, t)$. Since both $\mathbf{E}_s(z, t)$ and $\mathbf{P}_s(t)$ change slowly with respect to t ,

$$\left| \frac{\partial}{\partial t} \mathbf{P}_s(t) \right| \ll |\omega_s \mathbf{P}_s(t)|; \quad \left| \frac{\partial}{\partial t} \mathbf{E}_s(z, t) \right| \ll |\omega_s \mathbf{E}_s(z, t)| \quad (24)$$

we can approximate

$$\frac{\partial^2 \mathbf{E}_{\text{sig}}(z, t)}{\partial t^2} \approx -\omega_s^2 \mathbf{E}_s(z, t)e^{i(k'_s z - \omega_s t)} + c.c. \quad (25)$$

$$\frac{\partial^2 \mathbf{P}_{\text{NL}}(z, t)}{\partial t^2} \approx -\omega_s^2 \mathbf{P}_s(t)e^{i(k_s z - \omega_s t)} + c.c. \quad (26)$$

Similarly,

$$\frac{\partial^2 \mathbf{E}_{\text{sig}}(z, t)}{\partial z^2} = \left[-k_s'^2 \mathbf{E}_s(z, t) + 2ik'_s \frac{\partial \mathbf{E}_s(z, t)}{\partial z} + \frac{\partial^2 \mathbf{E}_s(z, t)}{\partial z^2} \right] e^{i(k'_s z - \omega_s t)} + c.c. \quad (27)$$

The first term cancels with $\frac{n_s^2}{c^2} \frac{\partial^2 \mathbf{E}_{\text{sig}}(z, t)}{\partial t^2}$. The last term can be ignored because $\mathbf{E}_s(z, t)$ changes slowly in space,

$$\left| \frac{\partial \mathbf{E}_s(z, t)}{\partial z} \right| \ll |k'_s \mathbf{E}_s(z, t)| \quad (28)$$

The final differential equation for $\mathbf{E}_s(z, t)$ is thus

$$2ik'_s \frac{\partial \mathbf{E}_s(z, t)}{\partial z} = -\frac{\omega_s^2}{\varepsilon_0 c^2} \mathbf{P}_s(t)e^{i\Delta k z} \quad (29)$$

Integration of this equation yields the expression of $\mathbf{E}_s(z, t)$. At $z = l$,

$$\boxed{\mathbf{E}_s(l, t) = i \frac{\omega_s l}{2n_s \varepsilon_0 c} \mathbf{P}_s(t) \operatorname{sinc}\left(\frac{\Delta k l}{2}\right) e^{i \frac{\Delta k l}{2}}} \quad (30)$$

where $\operatorname{sinc} x \equiv \frac{\sin x}{x}$. **The signal field is proportional to the oscillating nonlinear polarization field with a $\frac{\pi}{2}$ phase shift.** The intensity of the signal averaged over an optical period (measured at $z > l$) is thus

$$I_{\text{sig}}(t) = \frac{n_r c \varepsilon_0}{2} |\mathbf{E}_s(l, t)|^2 = \frac{\omega_s^2 l^2}{8n_s \varepsilon_0 c} |\mathbf{P}_s(t)|^2 \operatorname{sinc}^2\left(\frac{\Delta k l}{2}\right) \quad (31)$$

The strength of the signal is determined by the $\operatorname{sinc}(\Delta k l / 2)$ function. This function is largest when

$$\boxed{\Delta k l \ll \pi} \quad (32)$$

This is called the **phase matching condition**. Specifically, a perfect phase matching where $\Delta k = 0$ is equivalent to the momentum conservation condition for the photons. If the phase matching condition is not fulfilled, the fields generated at the different locations along the optical path length in the material interfere destructively thus diminish the signal.

The phase matching condition may also be derived from a QED perspective. In the quantum description of a $n + 1$ -wave mixing process in the frequency domain, the transition rate constant for n waves interact with one molecule and emit the signal is proportional to the square of the coupling

$$V^{(n+1)} \propto e^{i(\mathbf{k}_s - \sum' \mathbf{k}_j) \cdot \mathbf{r}} \left\langle f \left| \hat{\chi}^{(n)}(\omega_1, \omega_2, \dots, \omega_n) \right| i \right\rangle \cdot \mathbf{e}_1 \mathbf{e}_2 \dots \mathbf{e}_s \quad (33)$$

where $\sum'$ represents a certain linear combination of $\mathbf{k}_j$, $|i\rangle$ and $|f\rangle$ denotes the molecular states before and after the wave mixing process, $\hat{\chi}^{(n)}$ is an operator whose expectation value on a pure state is the n th-order hyperpolarizability of that state, and $\mathbf{e}_j$ and $\mathbf{e}_s$ are unit vectors along the directions of $\mathbf{E}_j$ and $\mathbf{E}_s$, respectively. To include the contribution of all molecules in a macroscopic sample, we need to integrate over $\mathbf{r}$. The oscillating $e^{i(\mathbf{k}_s - \sum' \mathbf{k}_j) \cdot \mathbf{r}}$ terms at different locations would cancel each other out except when $\mathbf{k}_s - \sum' \mathbf{k}_j = 0$. Mathematically, if the initial and final states of all molecules are the same throughout the sample, then an integration over the entire space leads to

$$\int V^{(n+1)} d\mathbf{r} \propto \delta(\mathbf{k}_s - \sum' \mathbf{k}_j) \left\langle f \left| \hat{\chi}^{(n)}(\omega_1, \omega_2, \dots, \omega_n) \right| i \right\rangle \cdot \mathbf{e}_1 \mathbf{e}_2 \dots \mathbf{e}_s \quad (34)$$

Therefore, we can only observe the signal in the direction $\mathbf{k}_s = \sum' \mathbf{k}_j$.

An important consequence of the phase matching condition is that we can separate the weak nonlinear signals from the much stronger linear signals and separate different nonlinear signals in a non-collinear setup, because they are propagating in different directions. For example, for three wave mixing, the sum frequency generation has the strongest signal in the direction $\mathbf{k}_1 + \mathbf{k}_2$, whereas the difference frequency signal is strongest in the direction $\mathbf{k}_1 - \mathbf{k}_2$. Both are different from the direction of the incoming waves, $\mathbf{k}_1$ and $\mathbf{k}_2$.

2 Nonlinear Response Theory and Liouville Space Pathways

From a microscopic perspective, the oscillation of the induced polarization $\mathbf{P}(t)^*$ is due to the electronic motion in the molecule under the perturbation of the incoming light,

$$\mathbf{P}(t) = \text{Tr} \left(\hat{U}^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}(t) \hat{\rho}_0 \right) \equiv \left\langle \hat{U}^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}(t) \right\rangle \quad (35)$$

where $\hat{\boldsymbol{\mu}}$ is the dipole operator of the molecule, $\hat{U}(t)$ is the time-evolution operator under the perturbation (interaction with the electric field). $\hat{\rho}_0$ is the equilibrium density operator without the perturbation. $\langle \dots \rangle \equiv \text{Tr}(\dots \hat{\rho}_0)$ denotes the ensemble average. Note that $\hat{\rho}_0$ appears on the right most side in the trace.

A perturbative expansion of $\hat{U}(t)$ with respect to the interaction $-\hat{\boldsymbol{\mu}} \cdot \mathbf{E}(t)$ leads to

$$\mathbf{P}(t) = \mathbf{P}_0 + \mathbf{P}^{(1)}(t) + \mathbf{P}^{(2)}(t) + \mathbf{P}^{(3)}(t) + \dots \quad (36)$$

where $\mathbf{P}_0 = \langle \hat{\boldsymbol{\mu}} \rangle$ is the permanent dipole of the molecule, and the n th-order terms is

$$\mathbf{P}^{(n)}(t) = \int_{-\infty}^{\infty} dt_n \dots \int_{-\infty}^{\infty} dt_1 \mathcal{R}^{(n)}(t_n, t_{n-1}, \dots, t_1) \cdot \underbrace{\mathbf{E}(t - t_n) \dots \mathbf{E}(t - t_n - \dots - t_1)}_{n \text{ times}} \quad (37)$$

The n -th order response function is defined as

$$\begin{aligned} \mathcal{R}^{(n)}(t_n, t_{n-1}, \dots, t_1) &= \left(\frac{i}{\hbar} \right)^n \theta(t_n) \theta(t_{n-1}) \dots \theta(t_1) \\ &\times \langle [[[\dots [\hat{\boldsymbol{\mu}}(t_n + \dots + t_1), \hat{\boldsymbol{\mu}}(t_{n-1} + \dots + t_1)] \dots], \hat{\boldsymbol{\mu}}(t_1)], \hat{\boldsymbol{\mu}}(0)] \rangle \end{aligned} \quad (38)$$

where $t_1, \dots, t_n$ are **time delays** and $\theta(t)$ is the step function. **In the n th-order response, the molecule interacts with the field n times. These interactions are separated by the time delays $t_1, \dots, t_n$, and the signal is measured at time t .** The intensity of the n th-order nonlinear signal is thus proportional to E^{2n} (Eq. (31)). When the field is weak, the nonlinear signals are much weaker than the linear response signals. Therefore, in order to observe nonlinear processes one must focus extremely large amounts of light on the material, which requires a laser. The nonlinear and linear signals can be separated using the phase matching condition.

The most widely used nonlinear technique are third-order nonlinear spectroscopies, such as the common pump-probe spectroscopy and 2D spectroscopy. The third-order response function is

$$\mathcal{R}^{(3)}(t_3, t_2, t_1) = \left(\frac{i}{\hbar} \right)^3 \theta(t_3) \theta(t_2) \theta(t_1) \langle [[[\hat{\boldsymbol{\mu}}(t_3 + t_2 + t_1), \hat{\boldsymbol{\mu}}(t_2 + t_1)], \hat{\boldsymbol{\mu}}(t_1)], \hat{\boldsymbol{\mu}}(0)] \rangle \quad (39)$$

The three commutators give $2^3 = 8$ terms, each represent a **Liouville space pathway**. Different Liouville pathways correspond to different physical processes due to the distinct interactions with

*We omitted the spatial dependence of $\mathbf{P}$ because the molecular size is generally much smaller than the wave length of the light.

external fields. We write them explicitly as

$$\begin{aligned}
\mathcal{R}^{(3)}(t_3, t_2, t_1) = & \left(\frac{i}{\hbar}\right)^3 \theta(t_3)\theta(t_2)\theta(t_1) \\
& \times \{ \langle \hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_2 + t_1) \hat{\mu}(t_1) \hat{\mu}(0) \rangle \\
& - \langle \hat{\mu}(t_2 + t_1) \hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_1) \hat{\mu}(0) \rangle \\
& - \langle \hat{\mu}(t_1) \hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_2 + t_1) \hat{\mu}(0) \rangle \\
& + \langle \hat{\mu}(t_1) \hat{\mu}(t_2 + t_1) \hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(0) \rangle \\
& - \langle \hat{\mu}(0) \hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_2 + t_1) \hat{\mu}(t_1) \rangle \\
& + \langle \hat{\mu}(0) \hat{\mu}(t_2 + t_1) \hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_1) \rangle \\
& + \langle \hat{\mu}(0) \hat{\mu}(t_1) \hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_2 + t_1) \rangle \\
& - \langle \hat{\mu}(0) \hat{\mu}(t_1) \hat{\mu}(t_2 + t_1) \hat{\mu}(t_3 + t_2 + t_1) \rangle \}
\end{aligned} \tag{40}$$

Again, note that in these ensemble averages, $\hat{\rho}_0$ appears on the right most side in the trace. Alternatively, we can rewrite these traces by **putting $\hat{\mu}(t_3 + t_2 + t_1)$ on the left most side. Action of $\hat{\mu}(t_3 + t_2 + t_1)$ on the density operator corresponds to the emission of the final signal.**

$$\begin{aligned}
\mathcal{R}^{(3)}(t_3, t_2, t_1) = & \left(\frac{i}{\hbar}\right)^3 \theta(t_3)\theta(t_2)\theta(t_1) \\
& \times \{ \text{Tr}(\hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_2 + t_1) \hat{\mu}(t_1) \hat{\mu}(0) \hat{\rho}_0) \\
& - \text{Tr}(\hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_1) \hat{\mu}(0) \hat{\rho}_0 \hat{\mu}(t_2 + t_1)) \\
& - \text{Tr}(\hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_2 + t_1) \hat{\mu}(0) \hat{\rho}_0 \hat{\mu}(t_1)) \\
& + \text{Tr}(\hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(0) \hat{\rho}_0 \hat{\mu}(t_1) \hat{\mu}(t_2 + t_1)) \\
& - \text{Tr}(\hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_2 + t_1) \hat{\mu}(t_1) \hat{\rho}_0 \hat{\mu}(0)) \\
& + \text{Tr}(\hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_1) \hat{\rho}_0 \hat{\mu}(0) \hat{\mu}(t_2 + t_1)) \\
& + \text{Tr}(\hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(t_2 + t_1) \hat{\rho}_0 \hat{\mu}(0) \hat{\mu}(t_1)) \\
& - \text{Tr}(\hat{\mu}(t_3 + t_2 + t_1) \hat{\rho}_0 \hat{\mu}(0) \hat{\mu}(t_1) \hat{\mu}(t_2 + t_1)) \}
\end{aligned} \tag{41}$$

The density matrix $\hat{\rho}_0 = \sum_a w_a |a\rangle\langle a|$ has a ket side and a bra side. Eq. (41) clearly shows which side of the density operator the dipole operator is acting on. For example, for the pathway

$$\text{Tr}(\hat{\mu}(t_3 + t_2 + t_1) \hat{\mu}(0) \hat{\rho}_0 \hat{\mu}(t_1) \hat{\mu}(t_2 + t_1))$$

the field first interact with the system from the ket side ($\hat{\mu}(0)$); after time delay t_1 , the field then interact with the system from the bra side ($\hat{\mu}(t_1)$); after another time delay t_2 , the field interact again with the system from the bra side ($\hat{\mu}(t_2 + t_1)$); finally, after a time delay t_3 , a signal is emitted from the ket side ($\hat{\mu}(t_3 + t_2 + t_1)$). We denote this term as “ket/bra/bra” interaction. Note that due to the invariance of trace under circular shift, signal emission from the ket side or from the bra side are equivalent. But conventionally, we put signal emission on the ket side.

Among the eight pathways, only four of them are independent, the rest corresponds to the complex conjugate of these four pathways. Specifically, the bra side interaction is the complex

conjugate of ket side interaction. For example, the “bra/ket/ket” interaction term is

$$\text{Tr}(\hat{\boldsymbol{\mu}}(t_3 + t_2 + t_1)\hat{\boldsymbol{\mu}}(t_2 + t_1)\hat{\boldsymbol{\mu}}(t_1)\hat{\rho}_0\hat{\boldsymbol{\mu}}(0))$$

It is easy to show that

$$[\text{Tr}(\hat{\boldsymbol{\mu}}(t_3 + t_2 + t_1)\hat{\boldsymbol{\mu}}(t_2 + t_1)\hat{\boldsymbol{\mu}}(t_1)\hat{\rho}_0\hat{\boldsymbol{\mu}}(0))]^* = \text{Tr}(\hat{\boldsymbol{\mu}}(t_3 + t_2 + t_1)\hat{\boldsymbol{\mu}}(0)\hat{\rho}_0\hat{\boldsymbol{\mu}}(t_1)\hat{\boldsymbol{\mu}}(t_2 + t_1))$$

Conventionally, we choose the four independent pathways to be the ket/ket/ket, bra/ket/bra, bra/bra/ket, and ket/bra/bra interactions.

$R_1 = \text{Tr}(\hat{\boldsymbol{\mu}}(t_3 + t_2 + t_1)\hat{\boldsymbol{\mu}}(t_2 + t_1)\hat{\boldsymbol{\mu}}(t_1)\hat{\boldsymbol{\mu}}(0)\hat{\rho}_0)$	ket/ket/ket	(42)
$R_2 = \text{Tr}(\hat{\boldsymbol{\mu}}(t_3 + t_2 + t_1)\hat{\boldsymbol{\mu}}(t_1)\hat{\rho}_0\hat{\boldsymbol{\mu}}(0)\hat{\boldsymbol{\mu}}(t_2 + t_1))$	bra/ket/bra	
$R_3 = \text{Tr}(\hat{\boldsymbol{\mu}}(t_3 + t_2 + t_1)\hat{\boldsymbol{\mu}}(t_2 + t_1)\hat{\rho}_0\hat{\boldsymbol{\mu}}(0)\hat{\boldsymbol{\mu}}(t_1))$	bra/bra/ket	
$R_4 = \text{Tr}(\hat{\boldsymbol{\mu}}(t_3 + t_2 + t_1)\hat{\boldsymbol{\mu}}(0)\hat{\rho}_0\hat{\boldsymbol{\mu}}(t_1)\hat{\boldsymbol{\mu}}(t_2 + t_1))$	ket/bra/bra	

3 Double-Sided Feynman Diagrams

The different terms in the nonlinear response function lead to an array of different nonlinear signals. Diagrammatic perturbation theory is a simplified way of keeping track of the contributions to a particular signal given a particular set of states. It uses a series of simple diagrams to represent the evolution of the density matrix and allows us to interpret the microscopic origin of a signal. We focus on the **double-sided Feynman diagrams**. They are the easiest way of tracking the state of coherences in different time periods, and for noting absorption and emission events.

In the nonlinear response function, each appearance of the dipole operator indicates an interaction with the field. For a plane wave given by $\mathbf{E}(t)e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)} + \mathbf{E}^*(t)e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$, we have two possible interaction, $\hat{\boldsymbol{\mu}} \cdot \mathbf{E}(t)e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$ and $\hat{\boldsymbol{\mu}} \cdot \mathbf{E}^*(t)e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$. In QED, a negative frequency is associated with the photon annihilation operator acting on the ket. Therefore, **action of $\hat{\boldsymbol{\mu}} \cdot \mathbf{E}(t)e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$ on the ket side corresponds to absorption (photon annihilation), whereas action of $\hat{\boldsymbol{\mu}} \cdot \mathbf{E}^*(t)e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$ on the ket side corresponds to stimulated emission (photon creation).** The interaction on the bra side is the complex conjugate of the interaction on the ket side. Thus, **action of $\hat{\boldsymbol{\mu}} \cdot \mathbf{E}(t)e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$ on the bra side corresponds to stimulated emission, whereas action of $\hat{\boldsymbol{\mu}} \cdot \mathbf{E}^*(t)e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$ on the bra side corresponds to absorption.**

3.1 Rules for Double-Sided Feynman Diagrams

The double-sided Feynman diagram can be created according to the following rules:

1. The vertical double lines represent the ket and bra sides of the density matrix.
2. Time evolution is upward.
3. Straight arrow intersecting the vertical lines represents interaction with the incoming field. Absorption is designated through an inward pointing arrow. Emission is an outward pointing

arrow. Action on the left line is action on the ket, whereas action on the right line is action on the bra.

4. System evolves freely under $\hat{H}_0$ between interactions. The density matrix element for that period is often explicitly written.
5. Emission of the signal is designated by a wavy arrow and is always from the ket side. Emission of the signal must return the system to a population state.
6. The wave vector and frequency of the signal is the sum of the wave vectors and frequencies of the incoming field represented by the straight arrows with appropriate signs. A right pointing arrow corresponds to a positive sign, whereas a left pointing arrow corresponds to a negative sign.

Equivalent to Rule 3, an arrow pointing to the right corresponds to the action of $\hat{\boldsymbol{\mu}} \cdot \mathbf{E}(t)e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$, whereas an arrow pointing to the left corresponds to the action of $\hat{\boldsymbol{\mu}} \cdot \mathbf{E}^*(t)e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$. Table 1 summarizes all possible interactions and their corresponding diagrammatic representations.

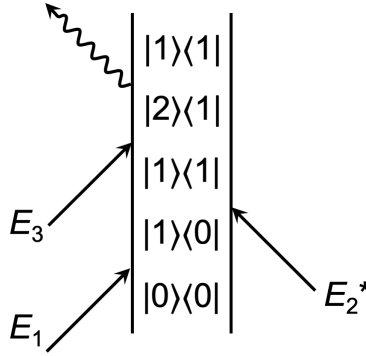
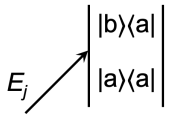
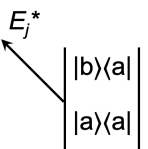
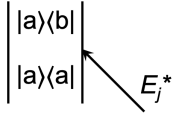
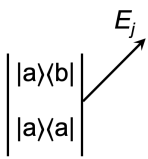
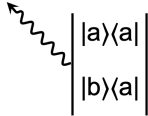


Figure 1: Example of Feynman diagram showing a ket/bra/ket interaction on a three-level system.

For example, the diagram shown in Figure 1 represent a ket/bra/ket interaction, corresponding to the pathway $-\text{Tr}(\hat{\boldsymbol{\mu}}(t_3 + t_2 + t_1)\hat{\boldsymbol{\mu}}(t_2 + t_1)\hat{\boldsymbol{\mu}}(0)\hat{\rho}_0\hat{\boldsymbol{\mu}}(t_1)) = -R_2^*$. Initially, the system stays on the ground state $\hat{\rho}_0 = |0\rangle\langle 0|$. Interaction with the first pulse $\mathbf{E}_1$ from the ket side generate a coherence ρ_{10} . This coherence evolve freely under $\hat{H}_0$ for time t_1 , until it interact with the second pulse $\mathbf{E}_2$ from the bra side. This interaction generates an excited population state $|1\rangle\langle 1|$. The system further evolve freely under $\hat{H}_0$ on this excited state for time t_2 , until it interact with the third pulse $\mathbf{E}_3$ from the ket side, generating another coherence ρ_{21} . This coherence then evolve freely under $\hat{H}_0$ for time t_3 , until a signal is emitted. The wave vector of the signal is $\mathbf{k}_s = \mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3$.

The advantage of diagrammatic perturbation theory is that these Feynman diagrams can be drawn intuitively for a given experimental setup (detection geometry, pulse sequence, etc.), and it is also straightforward to write down the mathematical expression of the response function corresponding to each diagram. This can be done by multiplying the Boltzmann population of the initial state, the matrix element of the dipole operator corresponding to each interaction, and the terms that describe the propagation under $\hat{H}_0$ between interactions. For this propagation, we need

Table 1: All Possible Interactions and Their Corresponding Diagrammatic Representations in a Double Sided Feynman Diagram

Interaction	Diagrammatic Repr.	Contrib. to $\mathcal{R}^{(n)}$	Contrib. to $\mathbf{k}_s$ and ω_s
Ket Side Absorption		$\boldsymbol{\mu}_{ba} \cdot \mathbf{E}_j$	$+\mathbf{k}_j, +\omega_j$
Ket Side Stimulated Emission		$\boldsymbol{\mu}_{ba} \cdot \mathbf{E}_j^*$	$-\mathbf{k}_j, -\omega_j$
Bra Side Absorption		$\boldsymbol{\mu}_{ab} \cdot \mathbf{E}_j^*$	$-\mathbf{k}_j, -\omega_j$
Bra Side Stimulated Emission		$\boldsymbol{\mu}_{ab} \cdot \mathbf{E}_j$	$+\mathbf{k}_j, +\omega_j$
Signal Emission		$\boldsymbol{\mu}_{ab} \cdot \mathbf{E}_s$	N/A

* $\boldsymbol{\mu}_{ba} = \langle b | \hat{\boldsymbol{\mu}} | a \rangle$ is the matrix element of the dipole operator.

to include the effects of relaxation of the system eigenstates due to system-bath coupling. The phenomenological equation of motion of the density matrix element is

$$\frac{d\rho_{ab}}{dt} = -i\omega_{ab}\rho_{ab} - \Gamma_{ab}\rho_{ab} \quad (43)$$

Therefore

$$\boxed{\rho_{ab}(t) = \rho_{ab}(0)e^{-i\omega_{ab}t - \Gamma_{ab}t}} \quad (44)$$

where $\omega_{ab} = \frac{E_a - E_b}{\hbar}$, E_a is the energy of state $|a\rangle$, Γ_{ab} is the rate constant of relaxation. **When $a = b$, $\Gamma_{aa} = 1/T_1$ corresponds to the population relaxation rate constant. When $a \neq b$, $\Gamma_{ab} = 1/T_2$ corresponds to the coherence dephasing rate constant.** Note that for coherence dephasing,

$$\Gamma_{ab} = \Gamma_{ba} \quad (45)$$

The expression of the response function can be written according to the following rules:

1. Start with a factor P_a signifying the probability of occupying the initial state, typically a Boltzmann factor.
2. Multiply by the matrix element of the dipole operator corresponding to each interaction with the incoming field and signal emission.
3. For the free evolution of population or coherence ρ_{ab} during time t , multiply the factor $e^{-i\omega_{ab}t - \Gamma_{ab}t}$.
4. Each diagram as an overall sign of $(-1)^m$ where m is the number of interactions from the bra side, because each time an interaction acts from the right in a commutator, it carries a minus sign.

For example, for the diagram shown in Figure 1, we have

$$-R_2^* = P_0 \mu_{10} \mu_{01} \mu_{21} \mu_{12} e^{-i\omega_{21}t_3 - \Gamma_{21}t_3} e^{-\Gamma_{11}t_2} e^{-i\omega_{10}t_1 - \Gamma_{10}t_1}$$

3.2 Third-Order Response of a Two-Level System

We will illustrate a practical use of the double-sided Feynman diagrams by calculating the third-order response function of a two-level system. We choose the frequency of the incoming waves to be similar to the transition energy of the two-level system. **Then by invoking the rotating wave approximation, we can eliminate certain diagrams, in which the photon is absorbed and the system is deexcited, or the photon is emitted and the system is excited.**

With the rotating wave approximation, for a two-level system, we can only have four diagrams, corresponding to R_1 through R_4 in Eq. (42). These diagrams are shown in Figure 2. From these diagrams we can immediately write down the expressions of R_1 through R_4 . After some rearrangement and using the relation $\Gamma_{10} = \Gamma_{01}$, $\omega_{10} = -\omega_{01}$, we have (from hereon we will treat the dipole moment as a scalar):

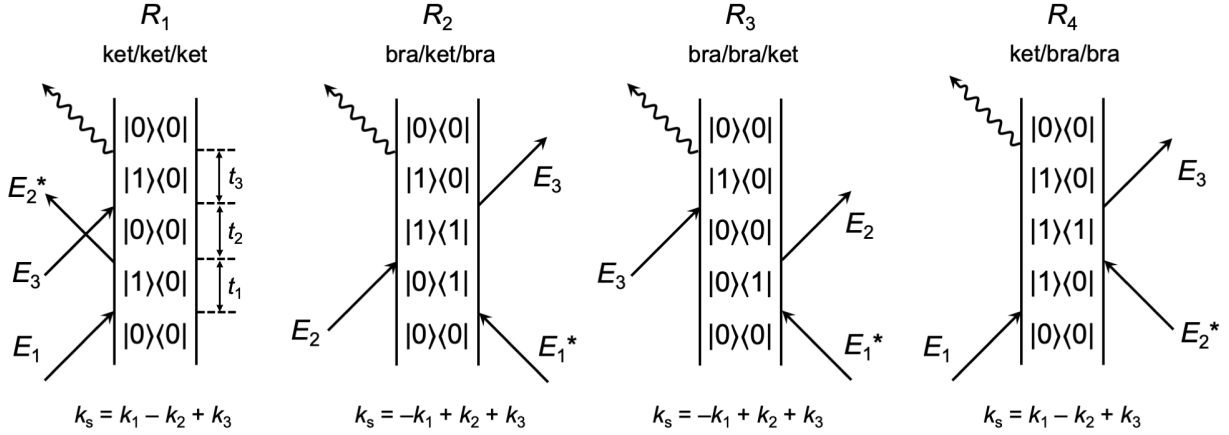


Figure 2: Possible Feynman diagrams for the third-order response of a two-level system.

$$\begin{aligned}
 R_1 &= P_0 |\mu_{10}|^4 \exp[-i\omega_{10}(t_3 + t_1) - \Gamma_{00}t_2 - \Gamma_{10}(t_1 + t_3)] \\
 R_2 &= P_0 |\mu_{10}|^4 \exp[-i\omega_{10}(t_3 - t_1) - \Gamma_{11}t_2 - \Gamma_{10}(t_1 + t_3)] \\
 R_3 &= P_0 |\mu_{10}|^4 \exp[-i\omega_{10}(t_3 - t_1) - \Gamma_{00}t_2 - \Gamma_{10}(t_1 + t_3)] \\
 R_4 &= P_0 |\mu_{10}|^4 \exp[-i\omega_{10}(t_3 + t_1) - \Gamma_{11}t_2 - \Gamma_{10}(t_1 + t_3)]
 \end{aligned} \tag{46}$$

For R_1 and R_4 , the coherence evolved during time t_1 is ρ_{10} . During this period, it acquires a phase $e^{-i\omega_{10}t_1}$. The same coherence continuously evolves during time t_3 , the phase acquired thus continuously grows. On the other hand, for R_2 and R_3 , the coherence evolved during time t_1 is ρ_{01} . During this period, it acquires a phase $e^{-i\omega_{01}t_1}$. During time t_3 , however, the conjugate coherence ρ_{10} is evolved. **Thus the phase acquired during t_1 is reversed in t_3 .** Therefore, R_2 and R_3 give the so-called **rephasing** signals, and R_1 and R_4 give **non-rephasing** signals. The rephasing signals are scattered in the direction $\mathbf{k}_2 - \mathbf{k}_1 + \mathbf{k}_3$, whereas the non-rephasing signals are scattered in the direction $\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3$.

4 Pump-Probe Spectroscopy

Lastly, we apply the diagrammatic method to study the pump-probe spectroscopy, and derive the expression of the familiar ground state bleach, stimulated emission, and excited state absorption signals. In a pump-probe experiment, we have two light pulses, the pump and the probe. The signal is detected in the same direction as the probe. The only combination of the incoming wave vectors resulting in this signal wave vector is $\mathbf{k}_s = \mathbf{k}_{\text{pu}} - \mathbf{k}_{\text{pu}} + \mathbf{k}_{\text{pr}}$, i.e., the system must interact with the pump pulse twice. Furthermore, we assume that both pump and probe pulses can be modeled as delta pulses, and the time separated the two pulses are much longer than the duration of the pulses. In this case, the first two interactions are with the pump pulse, and the duration

$t_1 \approx 0$, the third interaction is with the probe pulse. With these considerations, only six diagrams survives, as shown in Figure 3.

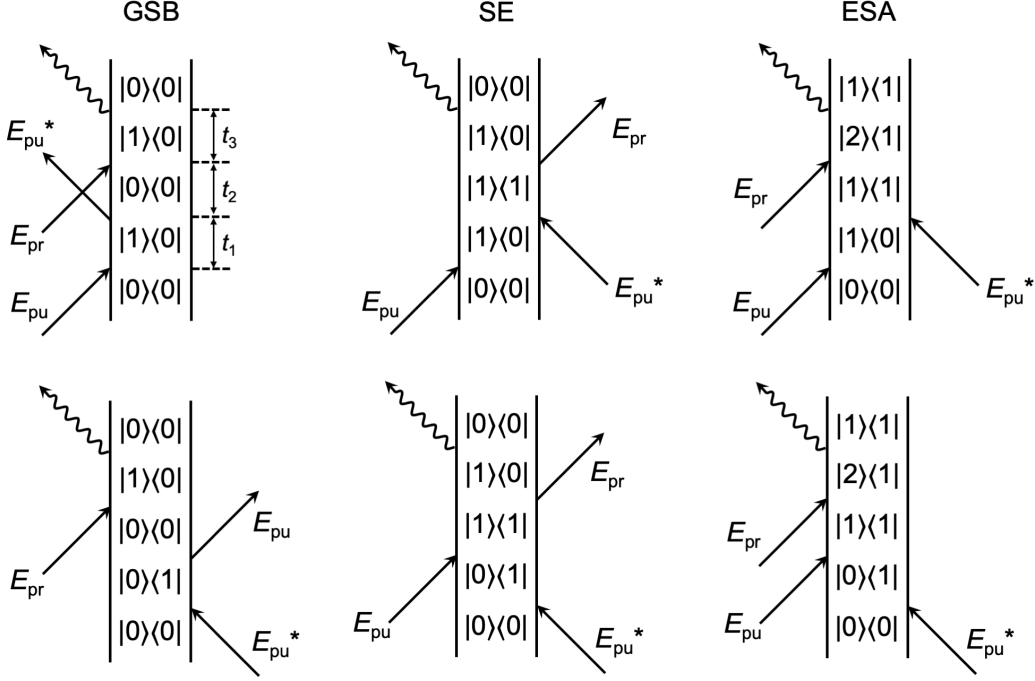


Figure 3: Possible Feynman diagrams for pump-probe spectroscopy.

The two diagrams in the left column describe those molecules which remain on the ground state after the pump pulse, and hence represent the ground state bleach (GSB) contribution. The middle diagrams are pumped to the excited state by the pump pulse and stimulated back to the ground state by the probe pulse. They represent stimulated emission (SE). The right diagrams are pumped to the excited state by the pump pulse, from where the second excited state is probed. They represent excited state absorption (ESA). The response functions of the GSB, SE, and ESA signals are

$$\begin{aligned}
 \mathcal{R}_{\text{GSB}}^{(3)}(t_3, t_2) &= -\frac{2i}{\hbar^3} \theta(t_3) \theta(t_2) P_0 |\mu_{10}|^4 e^{-\Gamma_{00} t_2} e^{-i\omega_{10} t_3 - \Gamma_{10} t_3} \\
 \mathcal{R}_{\text{SE}}^{(3)}(t_3, t_2) &= -\frac{2i}{\hbar^3} \theta(t_3) \theta(t_2) P_0 |\mu_{10}|^4 e^{-\Gamma_{11} t_2} e^{-i\omega_{10} t_3 - \Gamma_{10} t_3} \\
 \mathcal{R}_{\text{ESA}}^{(3)}(t_3, t_2) &= +\frac{2i}{\hbar^3} \theta(t_3) \theta(t_2) P_0 |\mu_{10}|^2 |\mu_{21}|^2 e^{-\Gamma_{11} t_2} e^{-i\omega_{21} t_3 - \Gamma_{21} t_3}
 \end{aligned} \tag{47}$$

Note that because $t_1 = 0$, in each column, the upper diagrams and the lower diagrams yield the same nonlinear response function. The ESA signal has an opposite sign compared to the GSB and SE signals, which originates from the fact that the ESA diagrams have 1, rather than 2 interactions from the bra side. The total response function is

$$\mathcal{R}^{(3)}(t_3, t_2) = \mathcal{R}_{\text{GSB}}^{(3)}(t_3, t_2) + \mathcal{R}_{\text{SE}}^{(3)}(t_3, t_2) + \mathcal{R}_{\text{ESA}}^{(3)}(t_3, t_2) \tag{48}$$

In general, when the incoming fields are impulsive, with τ_1 being the time delay between first and the second pulses and τ_2 being the time delay between the second and the last pulses, the envelopes of the incoming fields can be written as,

$$\begin{aligned} \mathbf{E}_1(t) &= \mathbf{E}_1\delta(t) \\ \mathbf{E}_2(t) &= \mathbf{E}_2\delta(t - \tau_1) \\ \mathbf{E}_3(t) &= \mathbf{E}_3\delta(t - \tau_1 - \tau_2) \end{aligned} \quad (49)$$

and the magnitude of the nonlinear polarization is simply

$$\boxed{P^{(3)}(t) \propto \mathcal{R}^{(3)}(t, \tau_2, \tau_1)} \quad (50)$$

For the case of pump-probe spectroscopy, $\tau_1 = 0$ and τ_2 is the time delay between the pump and probe pulses,

$$P^{(3)}(t) \propto \mathcal{R}^{(3)}(t, \tau_2) \quad (51)$$

In a pump-probe experiment, the signal is heterodyne detected. The field arrived at the detector is the superposition of the probe field and the signal field. The final result is often given by the absorbance difference spectrum $\Delta A(\omega)$, which is calculated from the ratio of field intensity arrived at the detector with and without the pump field present,

$$\Delta A(\omega) = -\log \frac{I_{\text{pumped}}(\omega)}{I_{\text{unpumped}}(\omega)} = -\log \frac{|E_{\text{pr}}(\omega) + E_{\text{sig}}(\omega)|^2}{|E_{\text{pr}}(\omega)|^2} \approx -\frac{2\text{Re}[E_{\text{pr}}^*(\omega)E_{\text{sig}}(\omega)]}{\ln(10)|E_{\text{pr}}(\omega)|^2} \quad (52)$$

According to Eqs. (30) and (51),

$$E_{\text{sig}}(t) \propto iP^{(3)}(t) \propto i\mathcal{R}^{(3)}(t, \tau_2) \quad (53)$$

Thus approximately,

$$\boxed{\Delta A(\omega) \propto -2\text{Re}E_{\text{sig}}(\omega) \propto 2\text{Im}\mathcal{R}^{(3)}(\omega, \tau_2)} \quad (54)$$

Fourier transformation of $\mathcal{R}^{(3)}(t, \tau_2)$ gives

$$\mathcal{R}^{(3)}(\omega, \tau_2) = \frac{2P_0}{\hbar^3} \left\{ \frac{|\mu_{10}|^4 e^{-\Gamma_{00}\tau_2}}{(\omega - \omega_{10}) + i\Gamma_{10}} + \frac{|\mu_{10}|^4 e^{-\Gamma_{11}\tau_2}}{(\omega - \omega_{10}) + i\Gamma_{10}} - \frac{|\mu_{10}|^2 |\mu_{21}|^2 e^{-\Gamma_{11}\tau_2}}{(\omega - \omega_{21}) + i\Gamma_{21}} \right\} \quad (55)$$

so

$$\boxed{2\text{Im}\mathcal{R}^{(3)}(\omega, \tau_2) = \frac{2P_0}{\hbar^3} \left\{ -\frac{2\Gamma_{10}|\mu_{10}|^4 e^{-\Gamma_{00}\tau_2}}{(\omega - \omega_{10})^2 + \Gamma_{10}^2} - \frac{2\Gamma_{10}|\mu_{10}|^4 e^{-\Gamma_{11}\tau_2}}{(\omega - \omega_{10})^2 + \Gamma_{10}^2} + \frac{2\Gamma_{21}|\mu_{10}|^2 |\mu_{21}|^2 e^{-\Gamma_{11}\tau_2}}{(\omega - \omega_{21})^2 + \Gamma_{21}^2} \right\}} \quad (56)$$

Hence, we will observe a negative Lorentzian line at the original frequency of the $|0\rangle \rightarrow |1\rangle$ transition (GSB and SE) and a positive Lorentzian line at the $|1\rangle \rightarrow |2\rangle$ transition (ESA). These signals decay exponentially with the time delay τ_2 between the pulses, due to the population relaxation during this time period.